# RL Infra/Ops Lab – Day 1: SSH hardening (GitHub-ready)

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**Što sam napravio (Day 1)**

- Podignuo Ubuntu Server VM u Hyper-V.
- Osigurao SSH pristup: ključ-only login (PasswordAuthentication OFF).
- Zabranio root login (PermitRootLogin no).
- Provjerio efektivne postavke preko `sshd -T` i testirao spajanje (FAIL bez ključa, OK s ključem).

**Okruženje**

- Host: Windows (PowerShell)
- VM: Ubuntu Server 24.04 LTS, hostname: infranode01
- User: opsuser
- IP (primjer iz lab-a): 172.21.27.74

**Windows PowerShell (komande)**

**Kreiraj .ssh folder (ako ne postoji)**

```
New-Item -ItemType Directory -Force $env:USERPROFILE\.ssh
```

**Generiraj ED25519 ključ**

```
ssh-keygen -t ed25519 -a 100 -f $env:USERPROFILE\.ssh\rl_lab_ed25519
```

**Provjeri da postoje ključ datoteke**

```
dir $env:USERPROFILE\.ssh\rl_lab_ed25519*
```

**Test spajanja (prvi put – prihvati host key)**

```
ssh opsuser@172.21.27.74
```

**Kopiraj public key na Ubuntu (dodaj u authorized_keys)**

```
type $env:USERPROFILE\.ssh\rl_lab_ed25519.pub | ssh opsuser@172.21.27.74
"mkdir -p ~/.ssh && chmod 700 ~/.ssh && cat >> ~/.ssh/authorized_keys &&
chmod 600 ~/.ssh/authorized_keys"
```

**Spajanje s ključem (mora raditi)**

```
ssh -i $env:USERPROFILE\.ssh\rl_lab_ed25519 opsuser@172.21.27.74
```

**Test nakon hardeninga bez ključa (mora FAIL)**

```
ssh opsuser@172.21.27.74
```

**Ubuntu (u SSH sesiji)**

**Provjeri status SSH servisa**

```
sudo systemctl status ssh --no-pager
```

**Omogući i odmah pokreni SSH servis (ako nije radio)**

```
sudo systemctl enable --now ssh
```

**Provjeri sluša li SSH na portu 22**

```
sudo ss -tulpen | grep ':22'
```

**Kreiraj override konfiguraciju (bez nano)**

```
sudo bash -c 'printf "\n# RL lab hardening\nPasswordAuthentication no\
nKbdInteractiveAuthentication no\n" > /etc/ssh/sshd_config.d/99-rl-
lab.conf'
```

**(Opcionalno) Final override ako nešto kasnije prepisuje postavke**

```
sudo bash -c 'cat >> /etc/ssh/sshd_config.d/99-rl-lab.conf << "EOF"\n\n#
Force-disable password auth (final override)\nMatch all\n
PasswordAuthentication no\n  KbdInteractiveAuthentication no\nEOF'
```

**Restart SSH da primijeni promjene**

```
sudo systemctl restart ssh
```

**Provjera efektivnih postavki**

```
sudo sshd -T | egrep 'passwordauthentication|kbdinteractiveauthentication|
permitrootlogin|pubkeyauthentication'
```

**Provjeri da override file postoji**

```
ls -l /etc/ssh/sshd_config.d/
```

**Pogledaj sadržaj override filea**

```
sudo cat /etc/ssh/sshd_config.d/99-rl-lab.conf
```

**Verifikacija**

**Bez ključa (Windows)**

```
ssh opsuser@172.21.27.74
```

- Očekivano: Permission denied (publickey).

**S ključem (Windows)**

```
ssh -i $env:USERPROFILE\.ssh\rl_lab_ed25519 opsuser@172.21.27.74
```

- Očekivano: login OK.

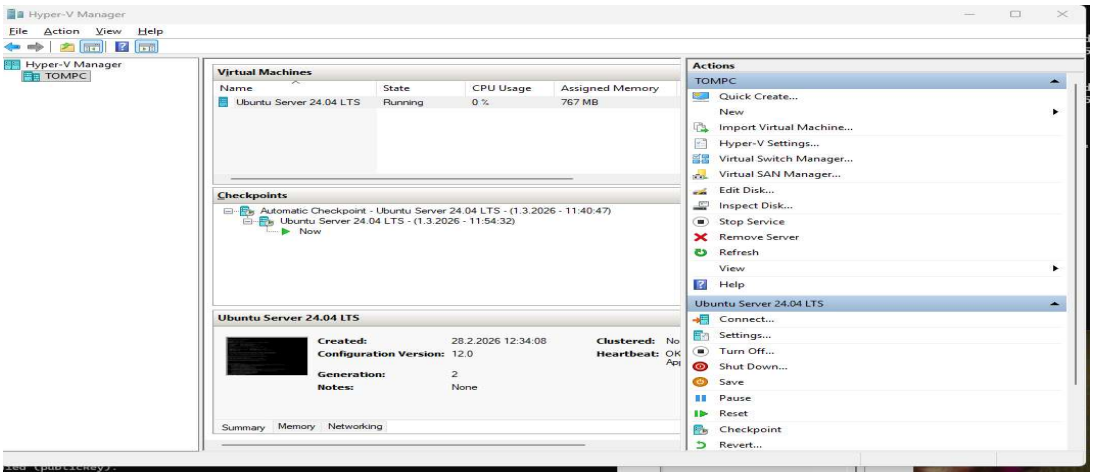
**Efektivni config (Ubuntu)**

```
sudo sshd -T | egrep 'passwordauthentication|kbdinteractiveauthentication|
permitrootlogin|pubkeyauthentication'
```

- Očekivano: passwordauthentication no, pubkeyauthentication yes, permitrootlogin no.

# Screenshotovi

*1) Hyper-V: VM je pokrenut (Running)*



*2) Test bez ključa: login je blokiran (Permission denied – publickey)*

```
Windows PowerShell
PS C:\Users\tomde> ssh opsuser@172.21.27.74
opsuser@172.21.27.74: Permission denied (publickey).
PS C:\Users\tomde>
```

*3) Test s ključem: login uspješan (opsuser@infranode01)*

```
opsuser@infranode01: ~
PS C:\Users\tomde> ssh -i $env:USERPROFILE\.ssh\rl_lab_ed25519 opsuser@172.21.27.74
Last login: Sat Feb 28 15:59:24 2026 from 172.21.16.1
opsuser@infranode01:~$
```

*4) Efektivna SSH konfiguracija (sshd -T): passwordauthentication no, permitrootlogin no*

```
opsuser@infranode01: ~
opsuser@infranode01:~$ sudo sshd -T | egrep "passwordauthentication|kbdinteractiveauthentication|permitrootlogin|pubkeya
uthentication"
permitrootlogin no
pubkeyauthentication yes
passwordauthentication no
kbdinteractiveauthentication no
opsuser@infranode01:~$
```